February 16, 2026 5:15 PM

Aldehydes and Ketones

↳ Both aldehydes and Ketones both contain the carbonyl group (C=O)

↳ In Aldehydes, the carbonyl group is attatched to the end carbon

↳ $R - \underset{\underset{O}{\|}}{C} - H$

↳ Ex: $\overset{4}{C}H_3 - \overset{3}{C}H_2 - \overset{2}{C}H_2 - \underset{\underset{O}{\|}}{\overset{1}{C}} - H$

1- butanol

Naming:

↳ ① Take the longest chain containing the carbonyl group, remove the "e" and add -al as the ending.

② The lowest number is given to the aldehyde

Ex: $\overset{2}{C}H_3 - \underset{\underset{O}{\|}}{\overset{1}{C}} - H$ Ethanal

↳ In Ketones, the carbonyl group is not attatched to the end carbon

↳ $R - \underset{\underset{O}{\|}}{C} - R'$

↳ Ex: $\overset{1}{C}H_3 - \underset{\underset{O}{\|}}{\overset{2}{C}} - \overset{3}{C}H_2 - \overset{4}{C}H_3$

2 - butanone

Naming:

↳ ① Take the longest chain containing the carbonyl group, remove the "e" and add -one as the ending.

② Indicate the position of the carbonyl using the lowest coefficient.

Ex: $\overset{6}{C}H_3 - \overset{5}{C}H_2 - \overset{4}{C}H_2 - \underset{\underset{O}{\|}}{\overset{3}{C}} - \overset{2}{C}H_2 - \overset{1}{C}H_3$

3- Hexanone

Properties:

↳ The carbonyl bond is polar

↳ Does not participate in hydrogen bonding.

↳ lower boiling point

↳ less soluble than other alcohols

↳ More soluble than hydrocarbons

↳ Can mix with polar and non-polar substances.

Reactions:

↳ **Oxidation** reactions: Alcohols → Aldehydes and Ketones.

↳ A gain of oxygen and loss of hydrogen (loss of electrons)

↳ **Oxidation reactions:** Alcohols → Aldehydes and Ketones.

↳ A gain of oxygen and loss of hydrogen (loss of electrons)

↳ <u>controlled</u> oxidation of alcohols, because complete oxidation of alcohols would result in the formation of CO_2 and H_2O

↳ Oxidation agents such as H_2O_2, $K_2Cr_2O_7$, and $KMnO_4$ are used in these reactions.

↳ Indicated by (O)

↳ (O) removes 2 H atoms, one from −OH group and one from the adjacent C atom. This results in a C=O group and a water molecule.

$$\underset{\text{alcohol}}{R-OH} + (O) \longrightarrow \underset{\text{aldehyde}}{R-\overset{O}{\overset{\|}{C}}H} \text{ or } R-\overset{O}{\overset{\|}{C}}-R' + H_2O$$

Primary alcohol oxidation: Aldehyde produced.

$$\underset{\text{ethanol } (1^\circ)}{CH_3-\overset{OH}{\overset{|}{C}H_2}} + (O) \longrightarrow \underset{\text{Ethanal}}{CH_3-\overset{O}{\overset{\|}{C}}H} + H_2O$$

Secondary alcohol oxidation: Ketone produced

$$\underset{\text{2-propanol } (2^\circ)}{CH_3-\overset{OH}{\overset{|}{C}H}-CH_3} + (O) \longrightarrow \underset{\text{2-propanone}}{CH_3-\overset{O}{\overset{\|}{C}}-CH_3} + H_2O$$

Tertiary alcohol oxidation: N-R

$$CH_3-\overset{CH_3}{\overset{|}{\underset{OH}{\underset{|}{C}}}}-CH_3 + (O) \longrightarrow N.R.$$

2-methyl-2-propanol (3°) [no H to grab]

↳ Hydrogenation Reaction: Aldehydes and Ketones ⟶ alcohols

↳ C=O double bond (in carbonyl group) can gain an addition reaction with H

↳ H appears at high temperatures [illegible] with a catalyst

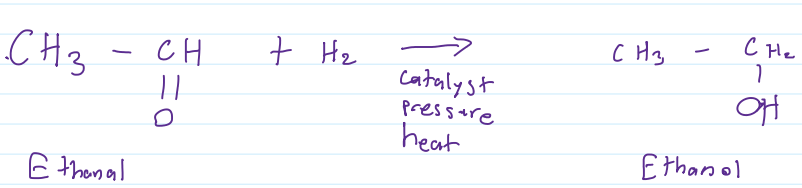
↳ C=O double bond (in carbonyl group) can gain an addition reaction with H

↳ Happens at high temperatures and pressure with a catalyst.

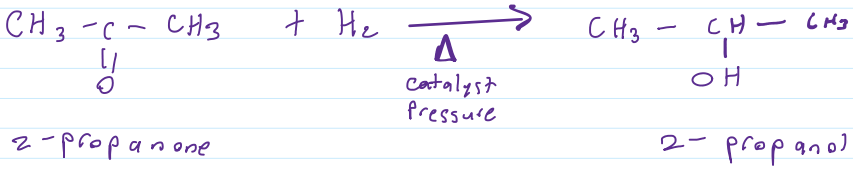
↳ opposite of oxidation and reduction **reaction**

↳ gain of hydration atoms / loss of oxygen atoms

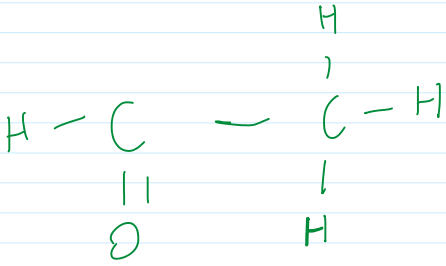
Ex:



Ethanal Ethanol



2-propanone 2-propanol

★ A compound with a C=O bond and the molecular formula C_2H_4O an aldehyde or ketone?

C_2H_4O is an Aldehyde